11-1c Industry Characteristics and Collusion vis-à-vis Competition

Given the benefits of collusion and the incentives to cheat, what industries are conducive to collusion vis-à-vis competition? Five factors emerge (Table 11.1). The first relevant factor is the number of firms or—more technically—the **concentration ratio** (The percentage of total industry sales accounted for by the top four, eight, or twenty firms.) , defined as the percentage of total industry sales accounted for by the top four, eight, or 20 firms. In general, the higher the concentration, the easier it is to organize collusion. Because the top four concentration in mobile wireless telecommunications services in the United States accounted for more than 90% of market share, the antitrust authorities blocked a merger of the second-largest firm, AT&T, with the fourth-largest firm, T-Mobile. The US Department of Justice argued:

Table 11.1

Industry Characteristics and Possibility of Collusion Vis-à-Vis Competition

Collusion possible	**Collusion difficult (competition likely)**
• Few firms (high concentration)	• Many firms (low concentration)
• Existence of an industry price leader	• No industry price leader
• Homogeneous products	• Heterogeneous products
• High barriers to entry	• Low barriers to entry
• High market commonality (mutual forbearance)	• Lack of market commonality (no mutual forbearance)

The substantial increase in concentration that would result from this merger, and the reduction in the number of nationwide providers from four to three, likely will lead to lessened competition due to an enhanced risk of anticompetitive coordination. Certain aspects of mobile wireless communications services markets, including transparent pricing, little buyer-side market power, and high barriers to entry and expansion, make them particularly conductive to coordination.

Second, the existence of a

price leader (A firm that has a dominant market share and sets “acceptable” prices and margins in the industry.)

—a firm that has a dominant market share and sets “acceptable” prices and margins in the industry—helps maintain order and stability needed for tacit collusion. The price leader can signal to the entire industry with its own pricing behavior, when it is appropriate to raise or reduce prices, without jeopardizing the overall industry structure. The price leader also possesses the

capacity to punish (Sufficient resources possessed by a price leader to deter and combat defection.)

, defined as sufficient resources to deter and combat defection. To combat cheating, the most frequently used punishment entails undercutting the defector by flooding the market with deep discounts, thus making the defection fruitless. Such punishment is costly because it will bring significant financial losses in the short run. However, if small-scale cheating is not addressed, defection may become endemic, and the price leader will have the most to lose if collusion collapses. Thus, a price leader such as De Beers must have both the willingness and the capacity to carry out punishments and bear the costs (see In Focus 11.1).

In Focus 11.1

Is a Diamond (Cartel) Forever?

The longest-running cartel in the modern world is the international diamond cartel headed by De Beers of South Africa. The cartel system underpinning the US$64 billion a year industry is, according to the *Economist*, “curious and anomalous—no other market exists, nor would anything similar be tolerated in a serious industry.”

A key reason diamonds were so expensive was because of the deeply ingrained perception of scarcity. If there was an oversupply, prices could plummet. Cecil Rhodes, an English tycoon who founded the De Beers Mines in South Africa in 1875, sought to solve this problem by focusing on two areas. First, Rhodes realized that supply from South Africa, the only significant producer in the world at that time, should be limited. Second, because producers (diggers) had little control over the quality and quantity of their output, they preferred to deal with an indiscriminate buyer willing to purchase both spectacular and mediocre stones. Since most output would be mediocre stones, producers preferred to remove any uncertainty and to be able to sell *all* of their output. On the other hand, buyers (merchants) preferred to secure a steady supply of stones (both high end and low end). Rhodes’ solution was to create an ongoing agreement between a single producer and a single buyer in which supply was kept low and prices high.

Putting his idea in action, Rhodes bought out all the major South African mines in the 1890s and formed a diamond merchants' association in the country, called the Diamond Syndicate, to which he would sell his output. In such "single-channel marketing," all members of the syndicate pledged to buy diamonds from Rhodes and sell them in specific quantities and prices. With such an explicit scheme of quantity-fixing and price-fixing, the diamond cartel was born.

Most cartels collapse due to organizational and incentive problems. The longevity of the De Beers cartel, running for more than 100 years, is very unusual. At least three attributes contribute to its longevity. First, the industry has an extraordinarily high concentration. In Rhodes' day, De Beers controlled all of South African (and hence virtually worldwide) production. Today, De Beers still controls approximately 40% of the world's rough diamond production, and its London-based wholly owned subsidiary Diamond Trading Company (DTC) sorts, values, and sells about 70% of the world's rough diamonds by value.

Carl Court/AFP/Getty Images

Second, De Beers is the undisputed price leader. Ten times a year, sales of rough diamonds (called "sights") are managed by the DTC to an exclusive group of cherry-picked "sightholders" from cities such as Antwerp, Johannesburg, Mumbai, New York, and Tel Aviv. Sightholders inform the DTC of their preferences, and the DTC then matchs them with inventory. During each sight, the DTC offers each sightholder a preselected parcel. The buyer either takes it or leaves it—no bargaining is permitted. Buyers usually take the parcel. If buyers repeatedly decline to take the parcel, they would not be invited again. This tactic allows De Beers to control, down to the carat, exactly what and how many stones enter the market and at what price. To maintain the exclusivity of the sightholders, their number has been decreased from about 350 in the 1970s to fewer than 100 in the 2000s.

Third, De Beers possesses both the willingness and the capability to enforce cartel arrangements. As in all cartels, the incentives to cheat are tremendous: Both producers and buyers are interested in cutting De Beers out of the process. As a price leader with a significant capacity to punish, De Beers's reactions are typically swift and powerful. In 1981, President Mobutu Seko of Zaire (now the Democratic Republic of the Congo) announced that his country would break away from De Beers by directly marketing its diamonds. Although only 3% of De Beers' sales were lost, its world order would be at stake if such actions went unpunished. Consequently, De Beers drew on its stockpiles to flood the market, driving the price of Zairian industrial diamonds from US$3 per carat to US$1.8 and wiping out any gains the Zairians hoped to grab. While incurring disproportional losses, De Beers made its point and Zaire crawled back on its knees.

Finally, De Beers faces one major institutional headache. The US government argued that De Beers was in clear violation of US antitrust laws and unsuccessfully tried to prosecute it in 1945, 1974, and 1994. De Beers managed to stay beyond the extraterritorial reach of US laws since it had no legal presence and no (direct) sales in the US. All its diamonds were sold in London, with sightholders then exporting them to the US, which is legal. However, with 50% of the retail diamond buyers in the US, these legal actions had prevented De Beers executives from being able to visit their buyers and retailers in the US for fear of being arrested. In 2008, De Beers settled the US charges for a total of US$295 million and agreed to operate in accordance with competition laws. A question thus looms large on the horizon for De Beers executives and antitrust officials: Has the longest-running cartel really come to an end? This truly is a billion-dollar question.

Sources: Based on *(1) Chicago Tribune, 2008, Diamond refunds are a consumer's best friend, January 21, www.chicagotribune.com; (2) A. Cockburn, 2002, Diamonds: The real story, National Geographic, March: 2–35; (3) Economist, 2004, The cartel isn't forever, July 17: 60–62; (4) Economist, 2011, Betting on De Beers, November 12: 73; (5) D. Spa, 1994, The Cooperative Edge: The Internal Politics of International Cartels, Ithaca, NY: Cornell University Press; (6) www.debeersgroup.com.*

Third, an industry with homogeneous products, in which rivals are forced to compete on price (rather than differentiation), is likely to lead to collusion. Because price competition is often "cut throat," firms may have stronger incentives to collude. Since the 1990s, many firms in commodity industries around the globe, such as ball bearings, car parts, shipping, and vitamins, have been convicted of price fixing.

Fourth, an industry with high entry barriers for new entrants (such as shipbuilding) is more likely to facilitate collusion than an industry with low entry barriers (such as restaurants). New entrants are likely to ignore the existing industry norms by introducing less homogeneous products with newer technologies (in other words, "disruptive technologies"). As "mavericks," new entrants can be loose cannons in otherwise tranquil industries. For example, SpaceX founder Elon Musk, who also founded Tesla, blasted the defense industry and the Pentagon for not using SpaceX's cheaper rockets. He sued the US Air Force in federal court for using a monopoly for rocket launch, United Launch Alliance, which is a joint venture between Boeing and Lockheed Martin. Incumbents naturally have collective interest in resisting such new entrants.

Finally, **market commonality** (The overlap between two rivals' markets.) —the degree of overlap between two rivals' markets—also has a significant bearing on the intensity of rivalry. A high degree of market commonality may restrain firms from aggressively going after each other.

Multimarket competition (Firms engage the same rivals in multiple markets.) occurs when firms engage the same rivals in multiple markets. Multimarket firms may respect their rivals' spheres of influence in certain markets and their rivals may reciprocate, leading to tacit collusion—an outcome known as **mutual forbearance** (Multimarket firms respect their rivals' spheres of influence in certain markets and their rivals reciprocate, leading to tacit collusion.) .

Mutual forbearance, due to a high degree of market commonality, primarily stems from two factors:

(1) deterrence and

(2) familiarity.

Deterrence is important because a high degree of market commonality suggests that if a firm attacks in one market, its rivals may engage in **cross-market retaliation** (Retaliatory attacks on a competitor's other markets if this competitor attacks a firm's original market.) , leading to a costly all-out war that nobody can afford. Familiarity is the extent to which tacit collusion is enhanced by a firm's awareness of the actions, intentions, and capabilities of rivals. Repeated interactions lead to such familiarity, resulting in more mutual respect. In the words of GE CEO Jeff Immelt:

GE has tremendous respect for traditional rivals like Siemens, Philips, and Rolls-Royce. But it knows how to compete with them; they will never destroy GE. By introducing products that create a new price-performance paradigm, however, the emerging giants [such as Mindry, Suzlon, Goldwind, and Haier] very well could.

Overall, the effectiveness of a firm's actions depends significantly on the domestic and international institutions governing competitive dynamics as well as firm-specific resources and capabilities. The next two sections expand on these points, which are illustrated in Figure 11.2.

Figure 11.2

Institutions, Resources, and Competitive Dynamics